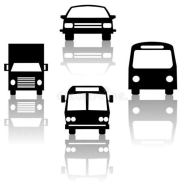


Vehicle Classification

vehicle .csv https://drive.google.com/drive/folders/1wYUP8fLY66PVcmnyIZTG45uJZsIHlmMk

vehicle_Adriana.ipynb

README_vehicle

Project Goal

This Machine Learning project focused on using Python to create a classification model “Prospect Auto” that would predict a vehicle class based on the silhouette.

List of Columns in DataFrame

- **`class`** column contains vehicle models. As we discussed previously, there may be 3 values there: `bus` , `van` or `car`.
- the rest of the columns are numerical columns that describe the **`silhouette`**(geometric features).
    - `compactness`
    - `circularity`
    - `distance_circularity`
    - `radius_ratio`
    - `pr.axis_aspect_ratio`
    - `max.length_aspect_ratio`
    - `scatter_ratio`
    - `elongatedness`
    - `pr.axis_rectangularity`
    - `max.length_rectangularity`
    - `scaled_variance`
    - `scaled_variance.1`
    - `scaled_radius_of_gyration`
    - `scaled_radius_of_gyration.1`
    - `skewness_about`
    - `skewness_about.1`
    - `skewness_about.2`
    - `hollows_ratio`

Steps in the Project

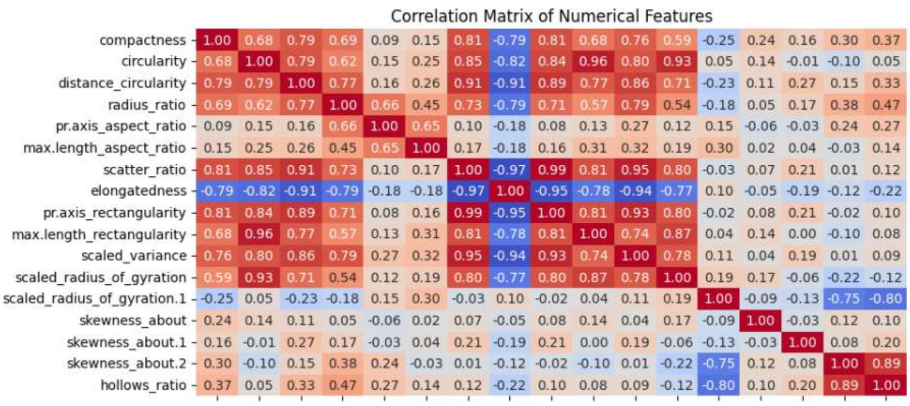
- **Import all of the libraries you need, upload the dataset and read it in.**

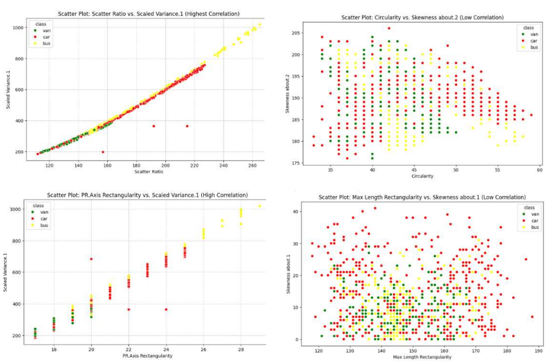
  - `import numpy as np`
  - `import pandas as pd`
  - `import matplotlib.pyplot as plt`
  - `import seaborn as sns`
  - `from sklearn.preprocessing import StandardScaler`
  - `from sklearn.model_selection import train_test_split`
  - `from sklearn.linear_model import LogisticRegression`
  - `from sklearn.ensemble import RandomForestClassifier`
  - `from sklearn.metrics import accuracy_score, confusion_matrix, classification_report, roc_auc_score`
  - `from sklearn.cluster import KMeans`
  - `from sklearn.metrics import silhouette_score`
  - `from sklearn.decomposition import PCA`

- **Prepare data & EDA**
  - **Initial Data Load**: The `vehicle.csv` dataset was loaded, containing 846 entries and 19 features.

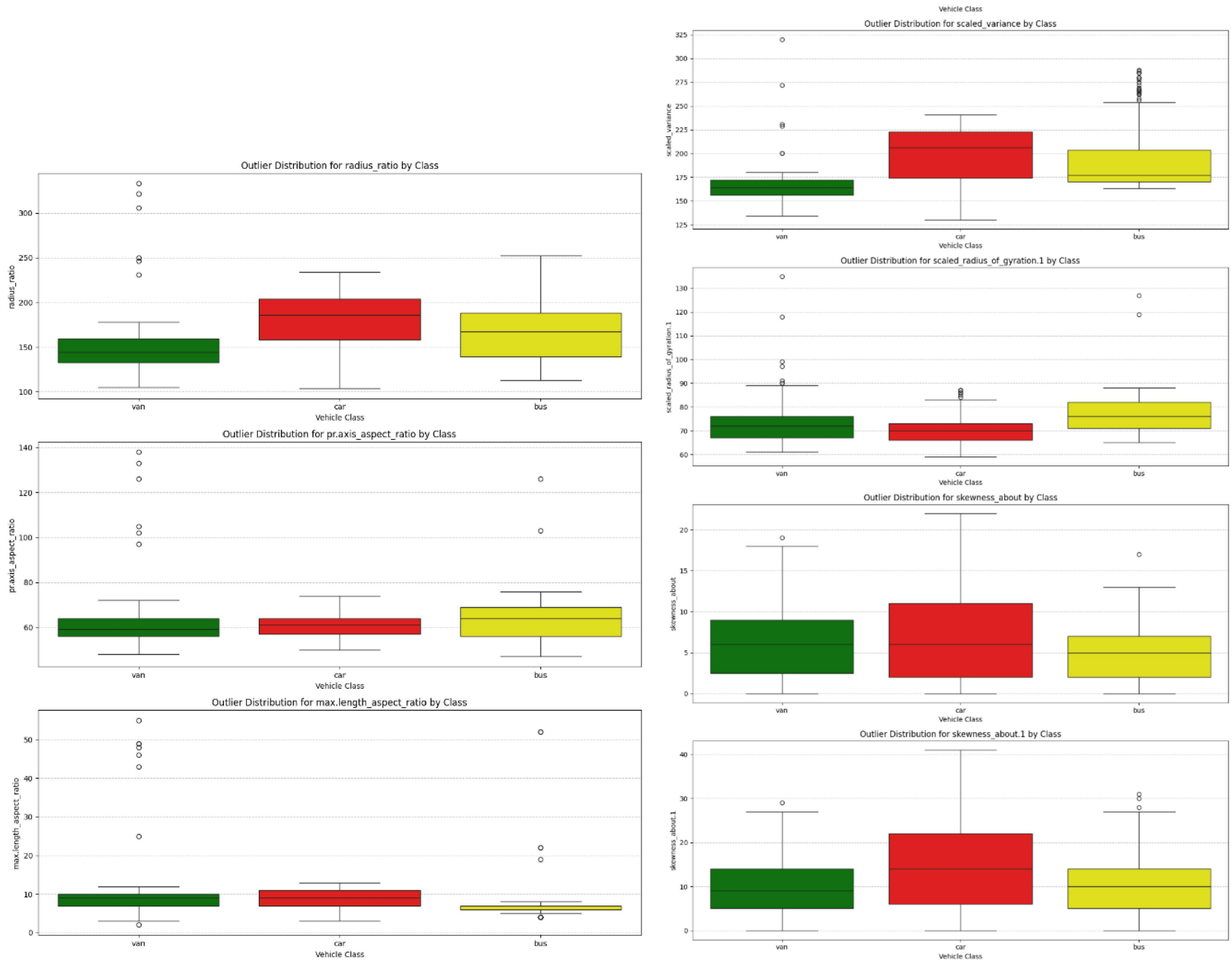
  Exploratory Data Analysis

  - **Missing Values**: 41 missing values identified across various numerical columns. These was imputed with the median of their respective columns.
  - **Feature Selection (Multicollinearity)**: An analysis of the correlation matrix revealed a very strong correlation (0.996) between `scatter_ratio` and `scaled_variance.1`. To reduce multicollinearity and simplify the model, `scaled_variance.1` was removed from the dataset.



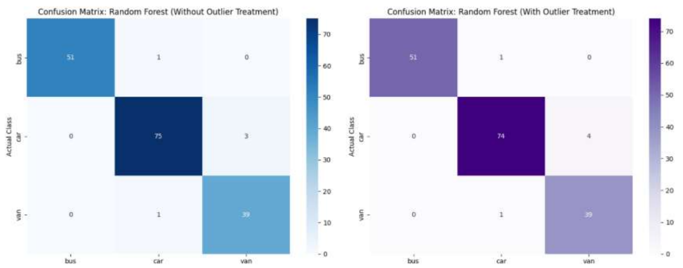


- **Outlier Detection**: Outliers were identified using the Interquartile Range (IQR) method in several numerical columns (`radius_ratio`, `pr.axis_aspect_ratio`, `max.length_aspect_ratio`, `scaled_variance`, `scaled_radius_of_gyration.1`, `skewness_about`, `skewness_about.1`).
    - **Relationship with Target Variable**: Box plots grouped by the 'class' variable showed that many identified 'outliers' were actually characteristic features of specific vehicle classes (e.g., certain `radius_ratio` values for 'van'). This suggested that these values might be discriminative rather than errors.
    - **Winsorization**: Winsorization (capping values at the 5th and 95th percentiles) was applied to the identified outlier columns to create a `df_treated` dataset.

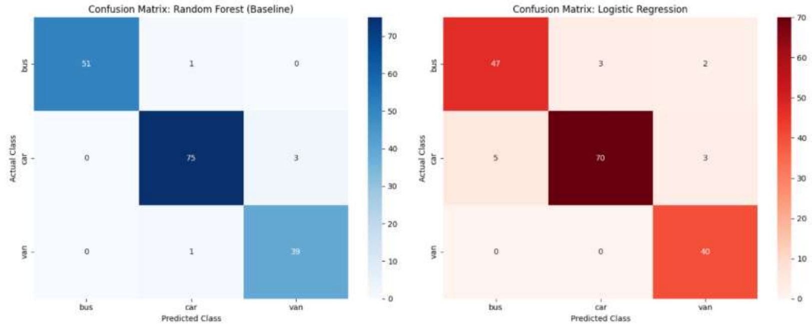


- **Prepare , train & test data** for **Supervised model**
    - `RandomForestClassifier` was trained on scaled data without outlier treatment.
    - `RandomForestClassifier` was trained on scaled data with outlier Winsorisation treatment
    - `LogisticRegression` model was trained on the same scaled data

- **Evaluate** the results



Model	Accuracy	F1-score (Macro Avg)
Random Forest (Without Outlier Treatment)	0.970588	0.969751
Random Forest (With Outlier Treatment)	0.964706	0.963696



Model	Accuracy	F1-score bus	F1-score car	F1-score van	ROC AUC bus	ROC AUC car	ROC AUC van
Random Forest (Baseline)	0.970588	0.990291	0.967742	0.951220	0.999674	0.996516	0.995865
Logistic Regression	0.923529	0.903846	0.927152	0.941176	0.981095	0.975892	0.998462

- ## Supervised Model : Conclusion , Comparison and Recommandation

Model	Accuracy	F1-score bus	F1-score car	F1-score van	ROC AUC bus	ROC AUC car	ROC AUC van
Random Forest (Baseline)	0.9706	0.9903	0.9677	0.9512	0.9993	0.9964	0.9954
Logistic Regression	0.9235	0.9038	0.9272	0.9412	0.9814	0.9760	0.9987

- **Random Forest Superiority**: The Random Forest (baseline) model consistently demonstrated superior performance compared to Logistic Regression across most metrics (Accuracy, F1-scores per class, and overall ROC AUC).
- **Robustness to Outliers**: The Random Forest model proved to be robust to outliers, as treating them with Winsorization did not improve its performance. This suggests that these 'outlier' values might contain valuable information for distinguishing vehicle classes.
- **Linear vs. Non-linear Models**: The performance difference highlights that the relationships within the dataset are likely complex and non-linear, making the tree-based Random Forest model more suitable than the linear Logistic Regression model.

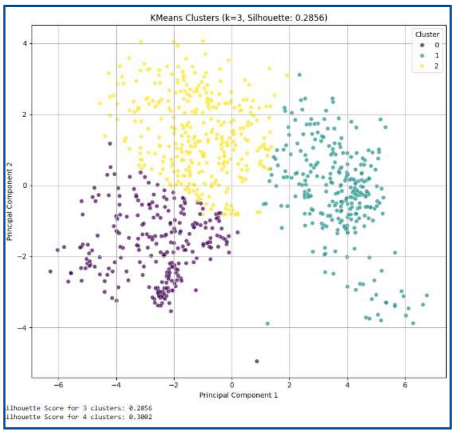
Recommendation: The **Random Forest Classifier trained without explicit outlier treatment** is recommended as the most effective and robust model for this vehicle classification task, offering excellent performance and good generalization.

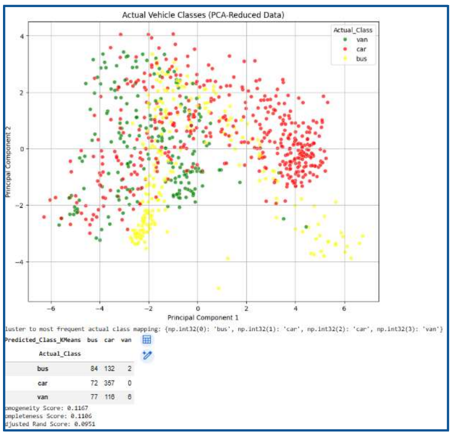
- **Data preparation** for **Unsupervised model**
  - **`KMeans`** `(Clustering)`

- separating the features from the target variable ('class')
- scaling the numerical features (`StandardScaler`)
- choose : number of clusters (`k`) using 'Elbow Method'
- use PCA to reduce dimensionality for visualization
- Evaluating KMeans Clustering after `Silhouette Score.` A higher Silhouette Score indicates better-defined clusters.

`KMeans` `(Clustering)`

- separating the features from the target variable ('class')
- scaling the numerical features (StandardScaler)
- choose : number of clusters (k) using 'Elbow Method' .





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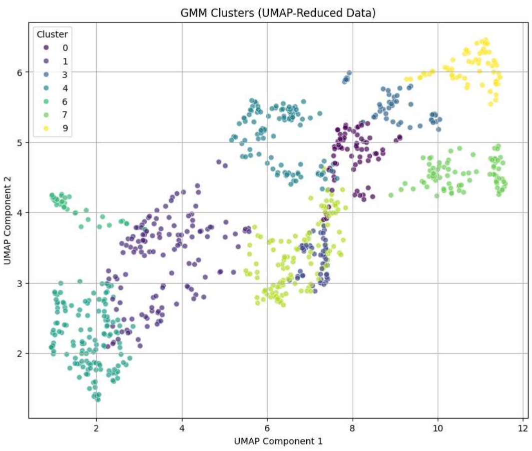
    ○ **GMM**(Clustering)

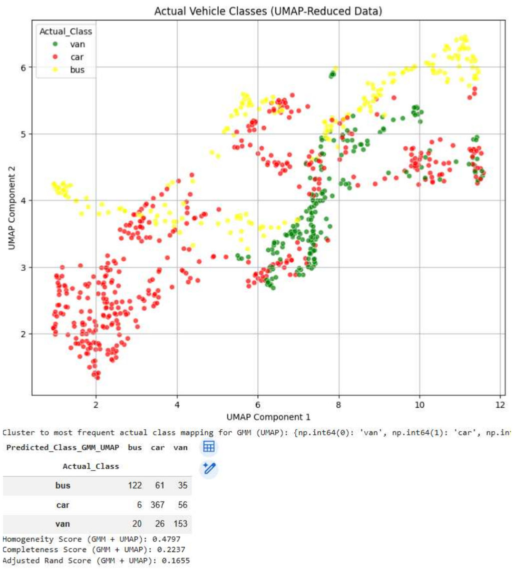
- from sklearn.mixture import GaussianMixture
- PCA for dimensionality reduction to 2 components for visualization
- GMM with the optimal number of components (e.g., optimal_n_components=3)

    ○ **GMM (UMAP)**

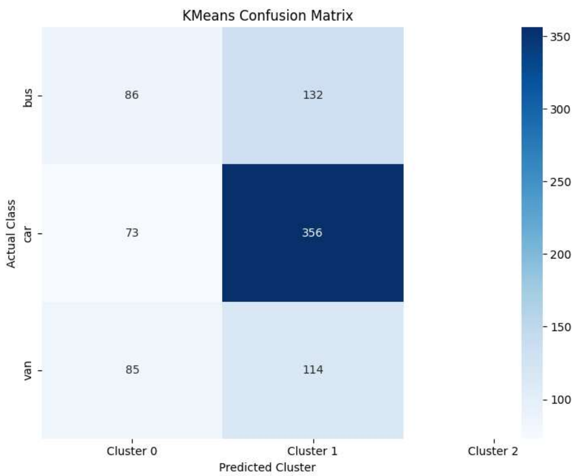
- from sklearn.mixture import GaussianMixture
- PCA for dimensionality reduction to 2 components for visualization
- GMM with the optimal number of components (e.g., optimal_n_components=3)

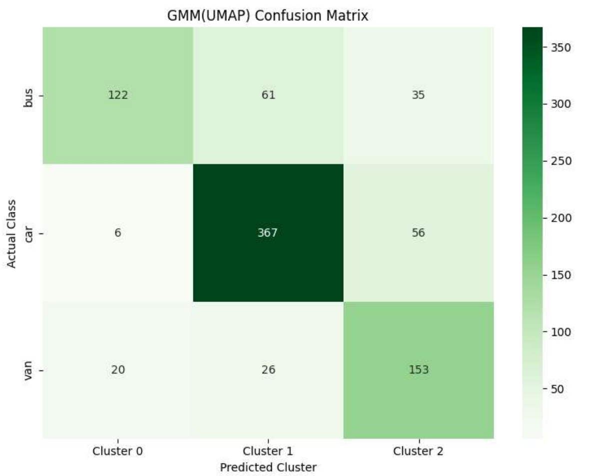
The average Silhouette Score for GMM clustering on UMAP data with 3 components is: 0.4320





- The Silhouette Scores, providing a comparison , are relatively very low, suggesting that the clusters formed by KMeans are not strongly separated or distinct.

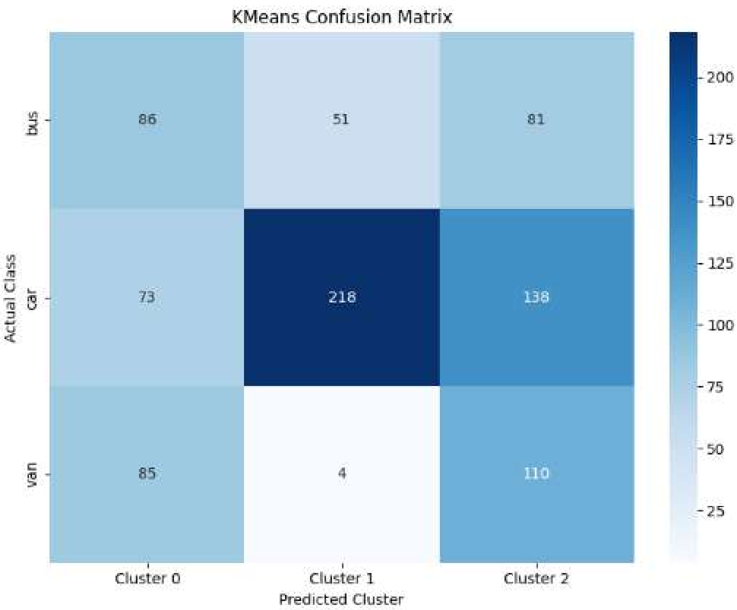


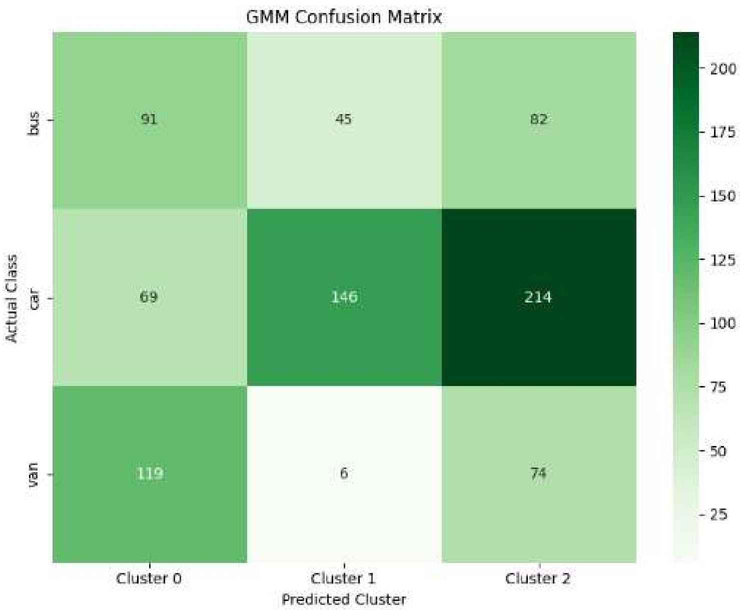


In essence, both KMeans and GMM(UMAP) demonstrated that the inherent structure of the data, when projected into a lower dimension (PCA) and subjected to unsupervised clustering with k=3 (corresponding to the number of actual classes), did not align perfectly with the true vehicle categories. **There was significant overlap and mixing of actual classes within the predicted clusters for both algorithms.**

- **Unsupervised Model : Conclusion , Comparison and Recommandation**

Model	Silhouette Score	Homogeneity Score	Completeness Score	Adjusted Rand Score
KMeans (k=3)	0.2856	0.1167	0.1106	0.0951
HAC (k=3)	0.2697	0.1225	0.1154	0.1076
DBSCAN (eps=1.5, min_samples=34)	0.0333	0.0757	0.3212	0.0656
GMM (n_components=3)	**0.3718**	**0.1554**	**0.2274**	0.0912





  - **KMeans** and **HAC** yielded similar, relatively low silhouette and purity scores.
  - **DBSCAN** performed poorly, identifying mostly noise, suggesting it was not well-suited for this dataset with the parameters chosen.
  - **Gaussian Mixture Models (GMM)** emerged as the best performer among the unsupervised models, achieving the highest Silhouette Score (0.3718) and better purity metrics (Homogeneity: 0.1554, Completeness: 0.2274) for 3 components.

Obs.: Cluster to Class Mapping: The mapping `cluster_to_class_mapping_gmm` indicated:

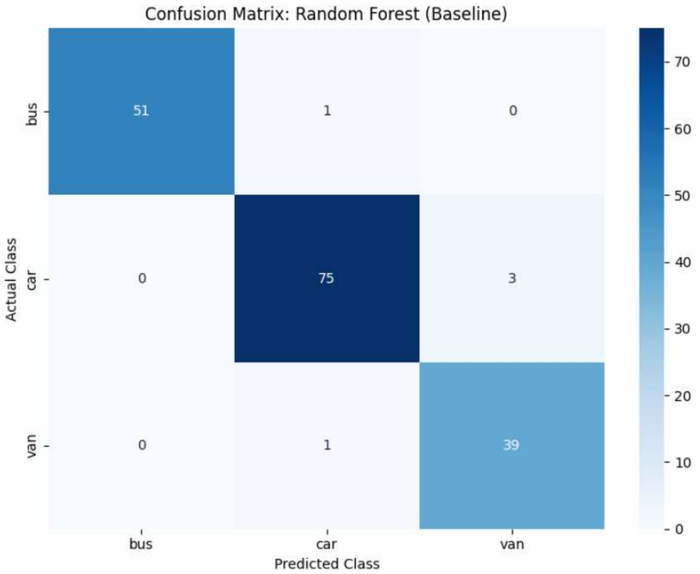
  - `GMM_Cluster 0` primarily corresponds to 'van'.
  - `GMM_Cluster 1` primarily corresponds to 'car'.
  - `GMM_Cluster 2` also primarily corresponds to 'car'. This already suggests that two of the GMM clusters are strongly associated with the 'car' class, which could lead to confusion or splitting of a single actual class into multiple clusters**.**

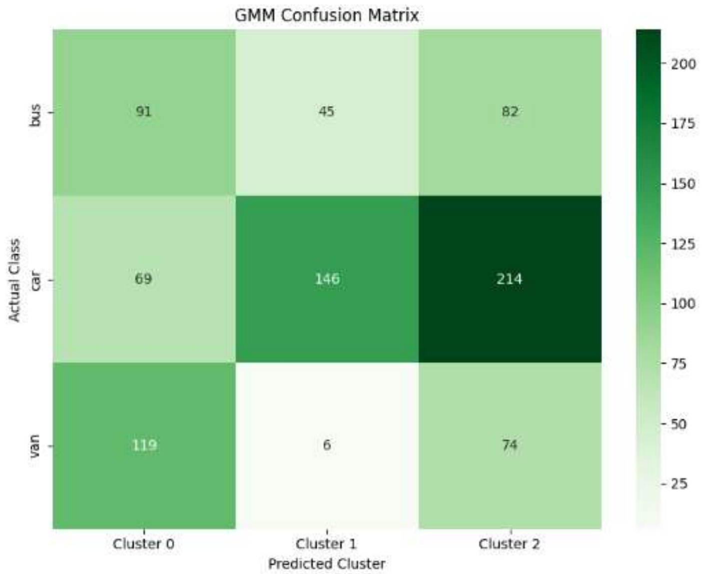
- **Supervised Models** vs. **Unsupervised GMM**

Direct Comparison: GMM vs. Supervised Models

Model	Type	Accuracy	F1-score bus	F1-score car	F1-score van	Homogeneity	Completeness	Adjusted Rand	Silhouette Score
Random Forest (Baseline)	Supervised	0.9706	0.9903	0.9677	0.9512	N/A	N/A	N/A	N/A
Logistic Regression	Supervised	0.9235	0.9038	0.9272	0.9412	N/A	N/A	N/A	N/A
GMM	Unsupervised	N/A	N/A	N/A	N/A	0.0963	0.0933	0.0734	0.2178

Confusion Matrix.   Actual Class Mapping vs. GMM

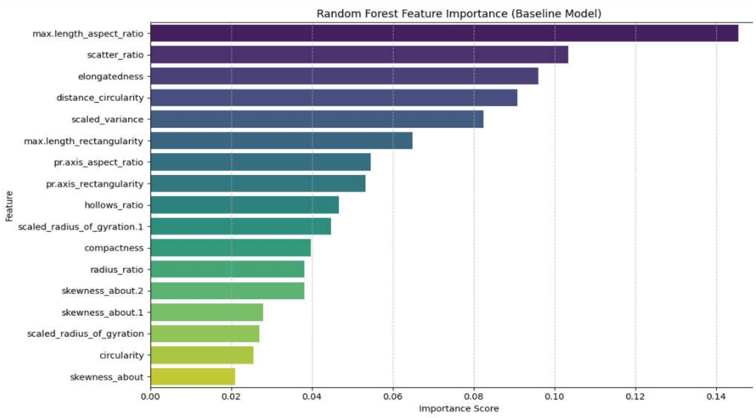


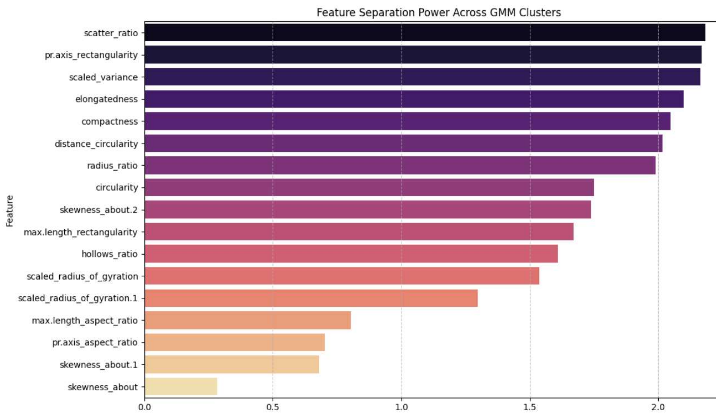


The matrix clearly shows significant mixing. For instance, 'bus' vehicles were split, with many incorrectly mapped to 'car' and 'van' clusters. Similarly, 'car' and 'van' instances were frequently misassigned to each other's mapped GMM clusters.

Feature Discriminatory Power:

- The feature importance analysis for Random Forest showed which features were crucial for this discrimination.
- In contrast, GMM, without explicit guidance from labels, struggled to group instances in a way that matches the actual classes.





Information Loss/Gain:

- Supervised models benefit from the explicit `class` label, allowing them to learn the boundaries between categories.
- Unsupervised models lack this information, and thus their clusters are based purely on feature similarity, which in this case, does not perfectly reflect the 'bus', 'car', 'van' categories.

Final report comparing the supervised and unsupervised models:

Direct Comparison: GMM vs. Supervised Models

Model	Type	Accuracy	F1-score bus	F1-score car	F1-score van	Homogeneity	Completeness	Adjusted Rand	Silhouette Score
Random Forest (Baseline)	Supervised	0.9706	0.9903	0.9677	0.9512	N/A	N/A	N/A	N/A
Logistic Regression	Supervised	0.9235	0.9038	0.9272	0.9412	N/A	N/A	N/A	N/A
GMM	Unsupervised	N/A	N/A	N/A	N/A	0.0963	0.0933	0.0734	0.2178

Recommendation

- For the task of vehicle classification on this dataset,

  Supervised Model ***Random Forest Classifier (without explicit outlier treatment)*** **is by far the superior model**.

  It demonstrates excellent predictive accuracy, F1-scores, and ROC AUC across all vehicle classes.

- **Unsupervised Model** ***Gaussian Mixture Models (GMM)*** useful for discovering hidden structures in unlabeled data, did not produce clusters that effectively align with the known vehicle classes.

  This suggests that the features, when analyzed in an unsupervised manner, do not inherently form distinct groups corresponding directly to the 'bus', 'car', and 'van' labels.

- Therefore, **if the goal is to classify vehicles into predefined categories,** ***a supervised approach is strongly recommended***.

  Unsupervised methods, in this context, serve more as an exploratory tool to understand inherent data groupings, rather than a direct solution for classification.